EPIC Meeting

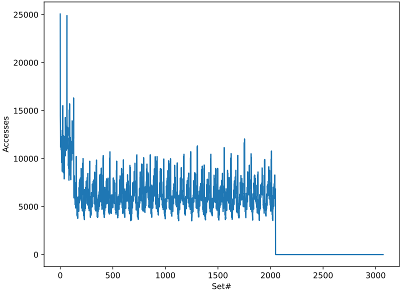
ChampSim, Set indexing Bug

Alexander V. Jamet   Dimitrios Chasapis   Georgios Vavouliotis
Marc Casas

Barcelona Supercomputing Center (BSC)

Content

Set indexing limitations

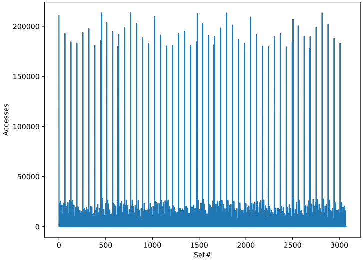


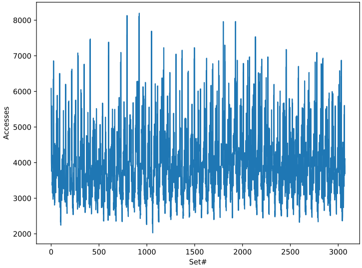
- Simplified version of indexing -> (address >> OFSSET_BITS) & bitmask(ln2(NUM_SETS))
  - Bitmask can only handle powers of two
- This is what actually hardware does

Solutions in simulators

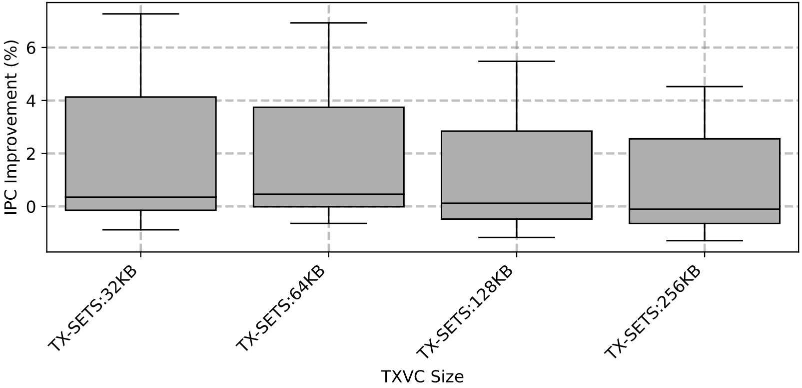
- Simulators often use different method to be able to compute indexes for arbitrary number of sets
  - Barrett / Granlund–Montgomery
- Tried the Barret method, but again noticed bad performance when comparing bitmask to barret version
  - Implementation of Barret resulted in very uneven distribution of addresses to sets. (see next slide)
- Implementation with just the mod operator provides better distribution, such as with the bitmask
  - Tried it and performance is good, which is the reason not to use it in a simualtor
  - The compiler actually does a similatr transformation to that of Barret

Barret vs Real Mod operator

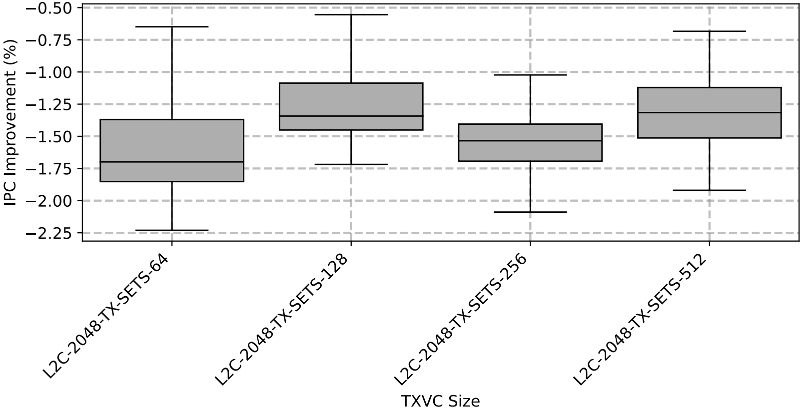




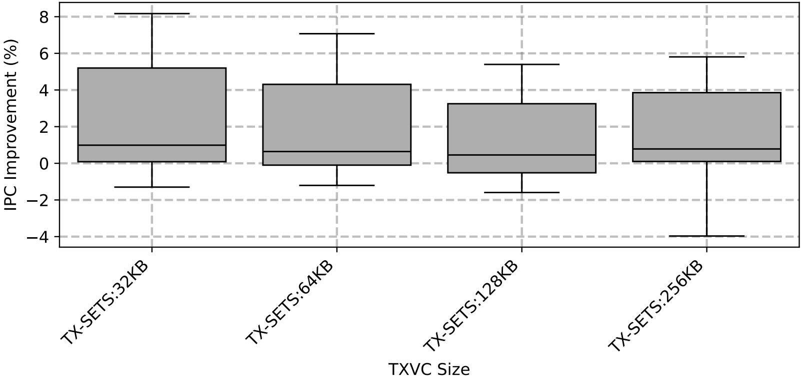
Performance of TX sets: L2C 1.5MB, LLC 1.5MB



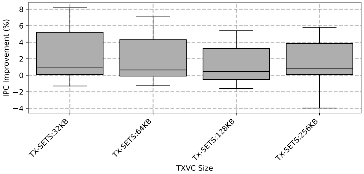
Performance of TX sets: L2C 2MB, LLC 1.5MB

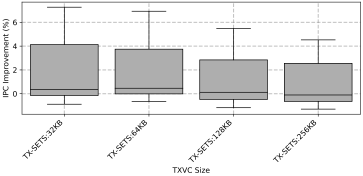


Performance of TX sets: L2C 1.5MB, LLC 1MB

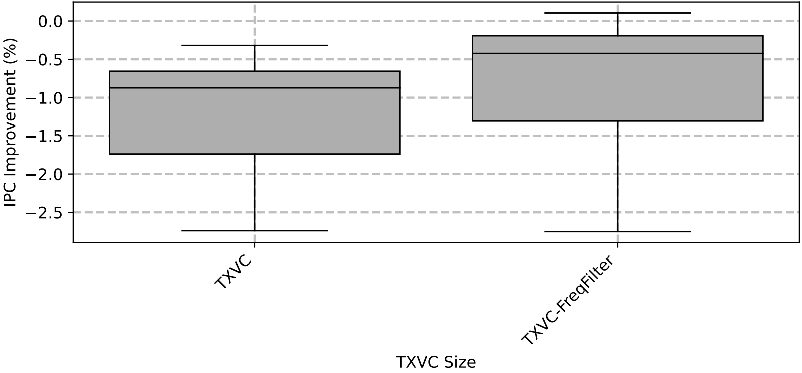


Compare results for LLC 1MB vs 1.5MB

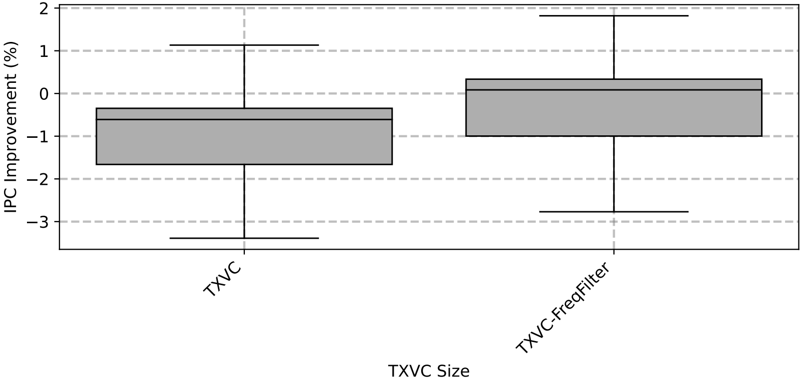




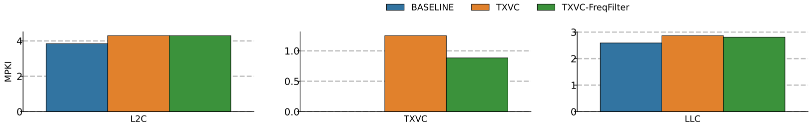
Performance of TXVC 64KB: L2C 2MB, LLC 1MB



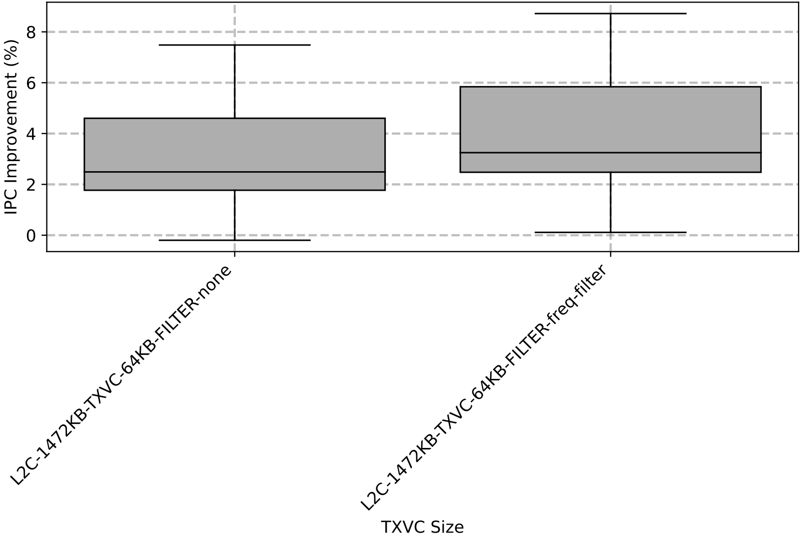
Performance of TXVC 128KB: L2C 2MB, LLC 1MB



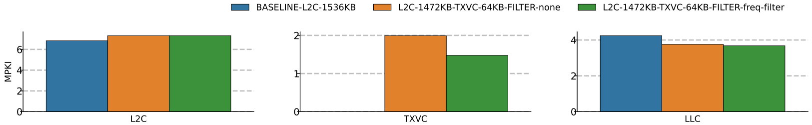
Performance of TXVC 128KB: L2C 2MB, LLC 1MB (MPKI)



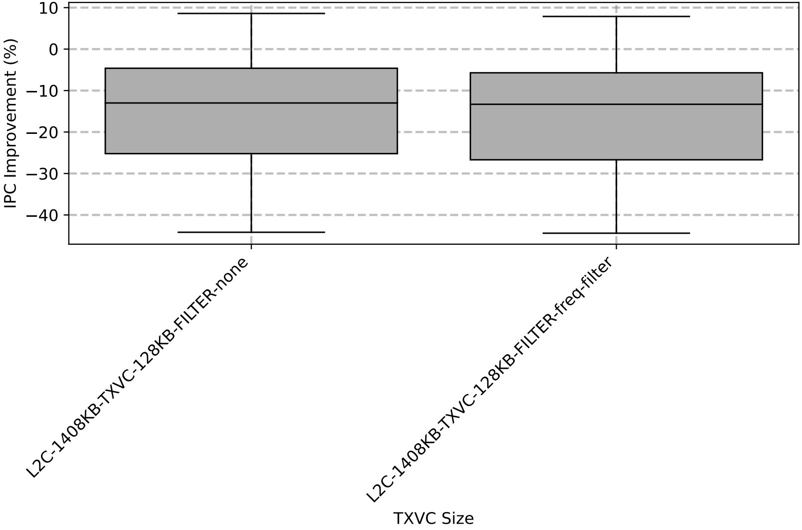
Performance of TXVC 64KB: L2C 1.5MB, LLC 1MB



Performance of TXVC 128KB: L2C 1.5MB, LLC 1MB (MPKI)



Performance of TXVC 128KB: L2C 1.5MB, LLC 1MB



Performance of TXVC 128KB: L2C 1.5MB, LLC 1MB (MPKI)